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Initial Steps - Part 1

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Before model building, you first need to perform the **test-train split** and **scale the features**.

Scaling of variables is an important step because, as you may have noticed, the variable 'area' is on a different scale with respect to all other numerical variables, which take very small values. Also, the categorical variables that you encoded earlier take either 0 or 1 as their values. Hence, it is important to have everything on the same scale for the model to be easily interpretable.

In the next video, you will learn the rescaling and splitting the dataset.



The screenshot shows a Jupyter Notebook interface. The top bar includes a file explorer, a menu bar (File, Edit, View, Insert, Cell, Kernel, Widgets, Help), and a status bar (Trusted, Python 3). The code cell contains the following Python code:

```
In [28]: df_train, df_test = train_test_split(housing, train_size=0.7, random_state=100)
print(df_train.shape)
print(df_test.shape)
```

The output of the code is:

```
(381, 14)
(164, 14)
```

The markdown cell below the code cell is titled "Rescaling the Features" and contains the following text:

As you saw in the demonstration for Simple Linear Regression, scaling doesn't impact your model. Here we can see that except for `area`, all the columns have small integer values. So it is extremely important to rescale the variables so that they have a comparable scale. If we don't have comparable scales, then some of the coefficients as obtained by fitting the regression model might be very large or very small as compared to the other coefficients. This might become very annoying at the time of model evaluation. So it is advised to use standardization or normalization so that the units of the coefficients obtained are all on the same scale. As you know, there are two common ways of rescaling:

1. Min-Max scaling
2. Standardisation (mean=0, sigma=1)

This time, we will use MinMax scaling.

You have seen the two popular rescaling methods- Min-Max scaling and Standardisation (mean=0 and sigma=1). The advantage of Standardisation over the other is that it doesn't compress the data between a particular range as in Min-Max scaling. This is useful, especially if there are extreme data point (outlier).

